

CH12 56

DATA
SET

DS1227

13. Geothermal Power Geothermal power is an important source of energy. Since the amount of energy contained in 1 pound of water is a function of its temperature, you might wonder whether water obtained from deeper wells contains more energy per pound. The data in the table are reproduced from an article on geothermal systems.¹⁷

Location of Well	Average (max.) Drill Hole Depth (m)	Average (max.) Temperature (°C)
El Tateo, Chile	650	230
Ahuachapan, El Salvador	1000	230
Namafjall, Iceland	1000	250
Larderello (region), Italy	600	200
Matsukawa, Japan	1000	220
Cerro Prieto, Mexico	800	300
Wairakei, New Zealand	800	230
Kizildere, Turkey	700	190
The Geysers, United States	1500	250

Is there a significant positive correlation between average maximum drill hole depth and average maximum temperature?

$$r = \frac{\sum xy}{\sqrt{\sum x^2 \sum y^2}} = \frac{23166.67}{\sqrt{6022.22 \times 8200}} = 0.3297$$

$$H_0: \rho = 0, H_a: \rho > 0$$

$$\alpha = 0.05$$

$$t_{STAT} = r \times \sqrt{\frac{n-2}{1-r^2}} \sim t_{n-2, 0.05}$$

$$t^* = 0.3297 \times \sqrt{\frac{9-2}{1-0.3297^2}} = 0.9240$$

$$RR: \{t \mid |t| > t_{n-2, 0.05}\} = \{t \mid |t| > 1.895\}$$

$$0.9240 < 1.895 \Rightarrow \text{not Reject } H_0$$

$\Rightarrow$ there's no significant positive correlation between

average maximum drill hole depth and average maximum temperature